Video Channel Presentation

Analyzing Youtube Video Performance Using Machine Learning

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Dataset Description

Dataset Name: YouTube Channel Performance Analytics

Source: Kaggle Dataset

Dataset contains 364 rows and 70 columns, with each row representing an individual YouTube video.

Variables are organized into the following categories:

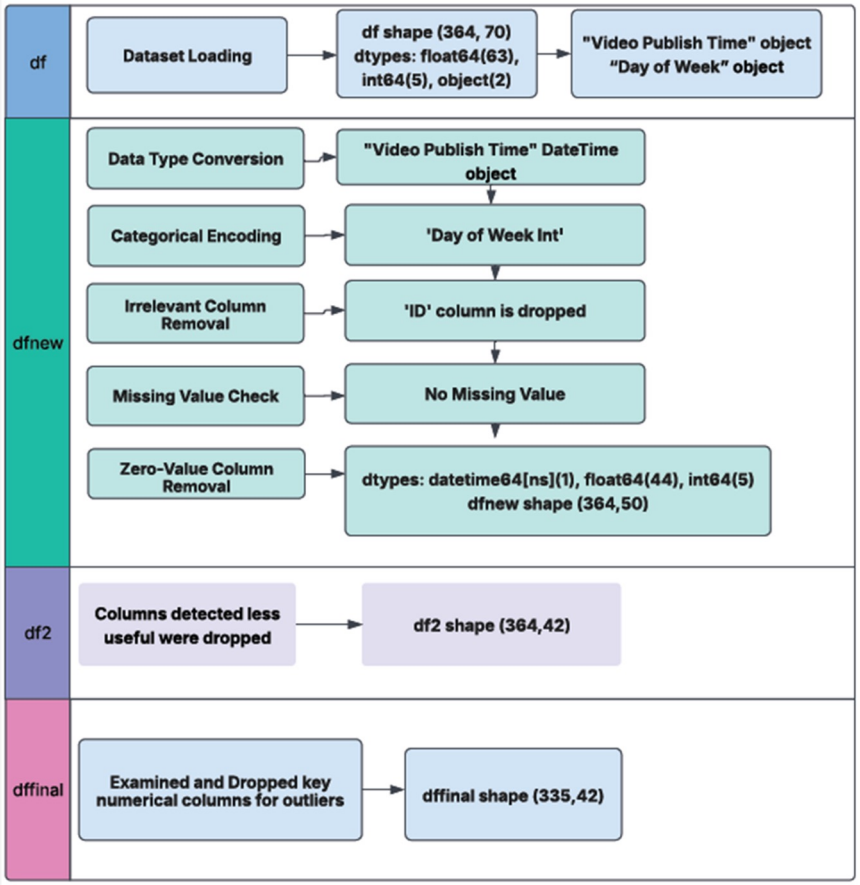
- Video Performance & Viewer Engagement Variable Description
- Video Interactivity Variable Description
- Monetization & Revenue Variable Description
- End Screen and Teaser Interactions Variable Description
- Video Metadata Variable Description

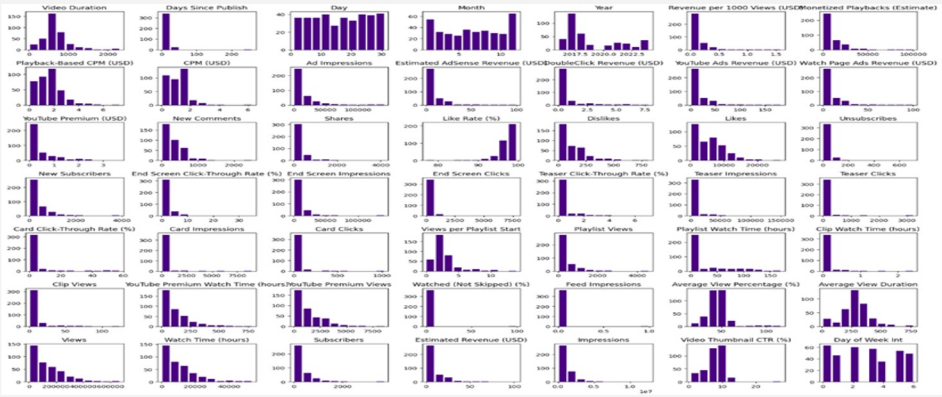
Analysis Goals

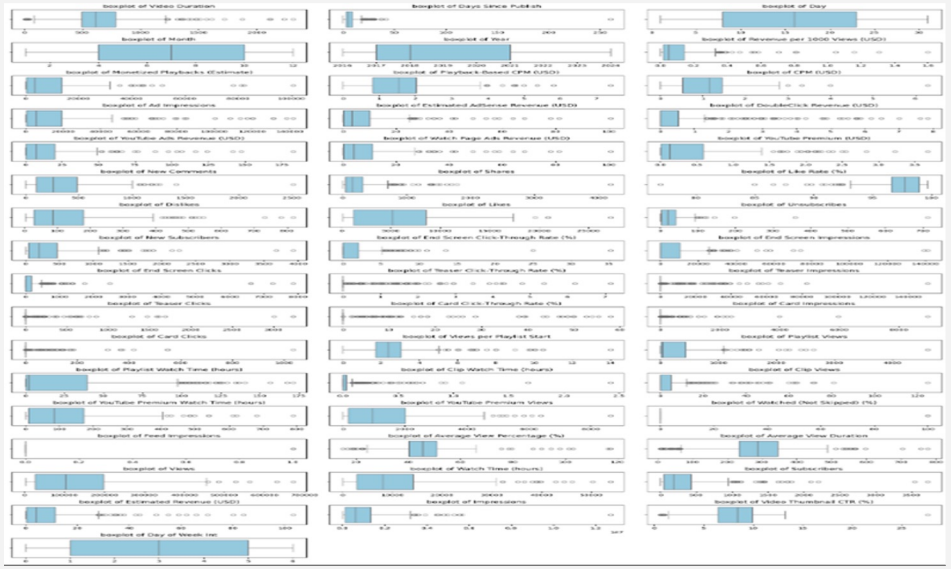
Research Questions

1. What video characteristics influence the average view percentage of YouTube videos?
2. What factors influence the revenue per 1000 views generated by YouTube videos?
3. What factors influence the return of viewers and overall retention rates?

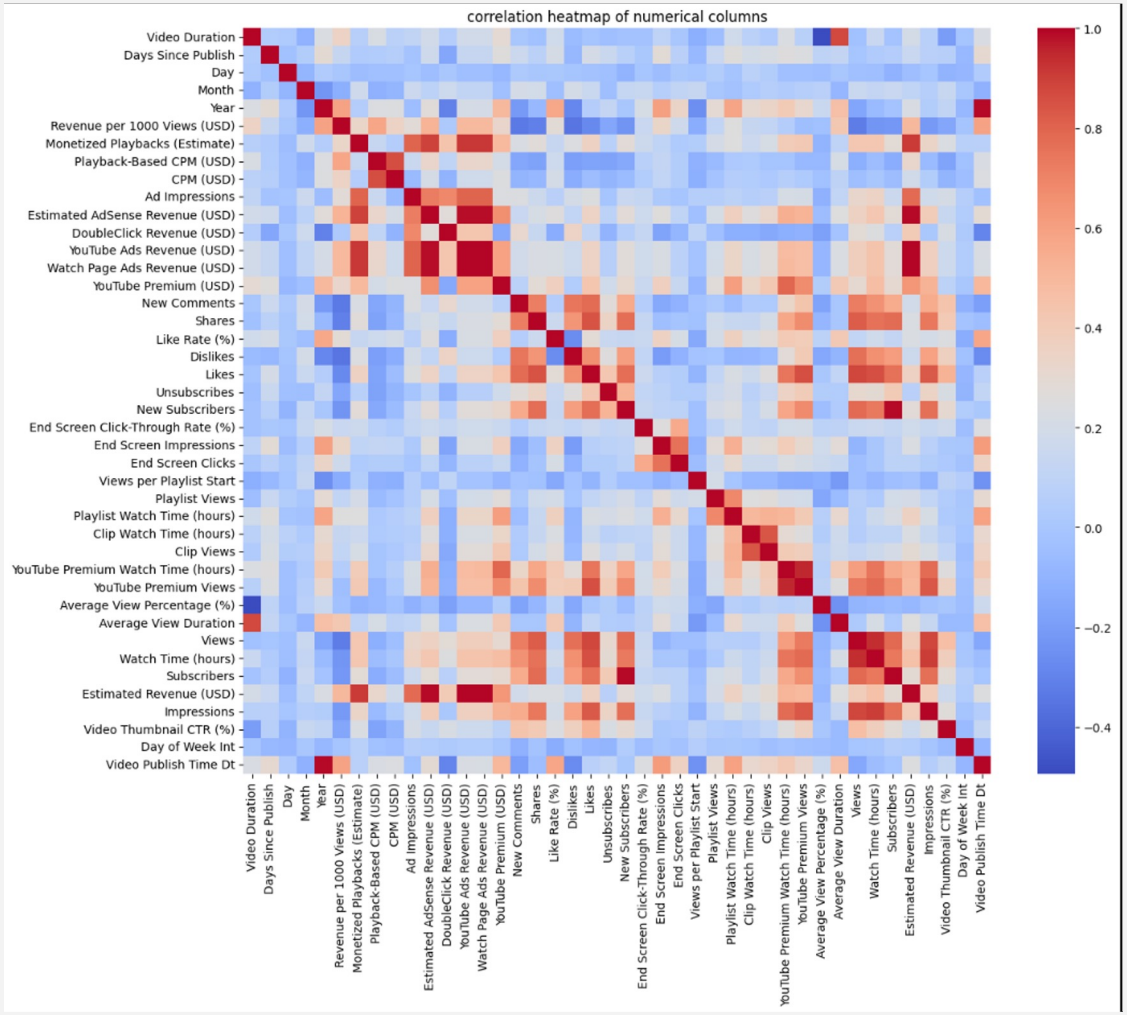
Data Cleaning Workflow







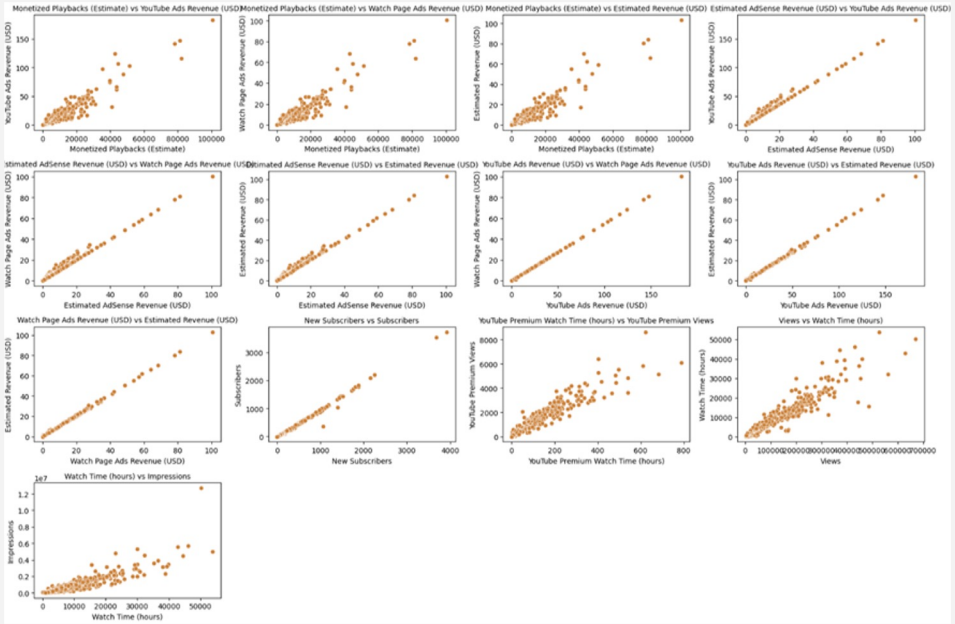
Exploratory Data Analysis- Overall



Exploratory Data Analysis- Overall

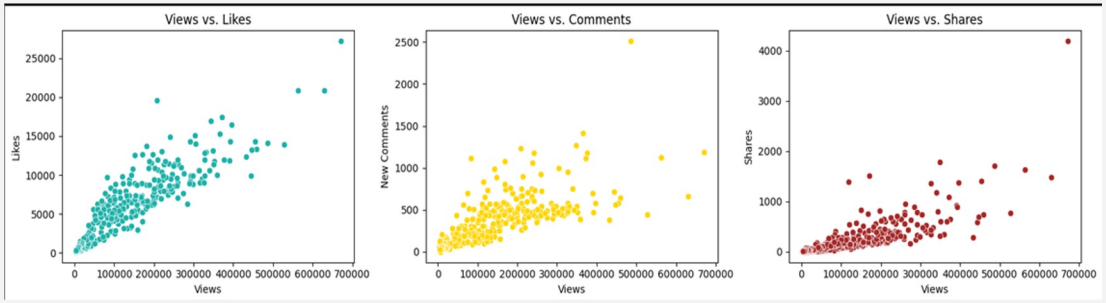
Strongly correlated pairs from the matrix

	Variable 1	Variable 2	Correlation
0	Monetized Playbacks (Estimate)	YouTube Ads Revenue (USD)	0.912912
1	Monetized Playbacks (Estimate)	Watch Page Ads Revenue (USD)	0.912978
2	Monetized Playbacks (Estimate)	Estimated Revenue (USD)	0.908697
3	Estimated AdSense Revenue (USD)	YouTube Ads Revenue (USD)	0.986704
4	Estimated AdSense Revenue (USD)	Watch Page Ads Revenue (USD)	0.986754
5	Estimated AdSense Revenue (USD)	Estimated Revenue (USD)	0.989912
6	YouTube Ads Revenue (USD)	Watch Page Ads Revenue (USD)	0.999938
7	YouTube Ads Revenue (USD)	Estimated Revenue (USD)	0.998853
8	Watch Page Ads Revenue (USD)	Estimated Revenue (USD)	0.998908
9	New Subscribers	Subscribers	0.991359
10	YouTube Premium Watch Time (hours)	YouTube Premium Views	0.948599
11	Views	Watch Time (hours)	0.940379
12	Watch Time (hours)	Impressions	0.901090

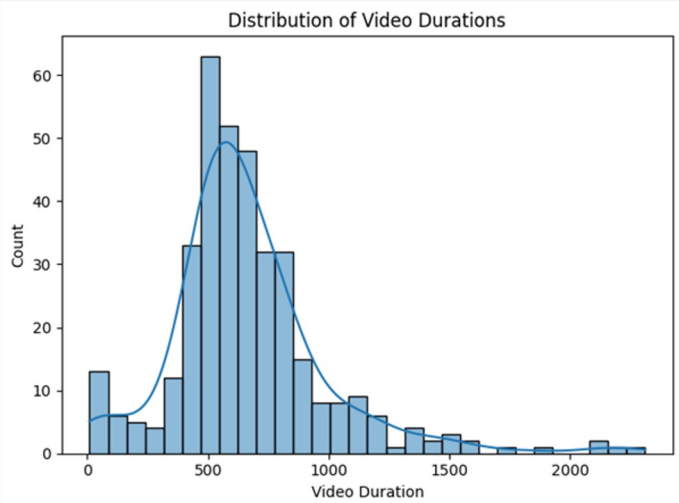


Exploratory Data Analysis- Individual

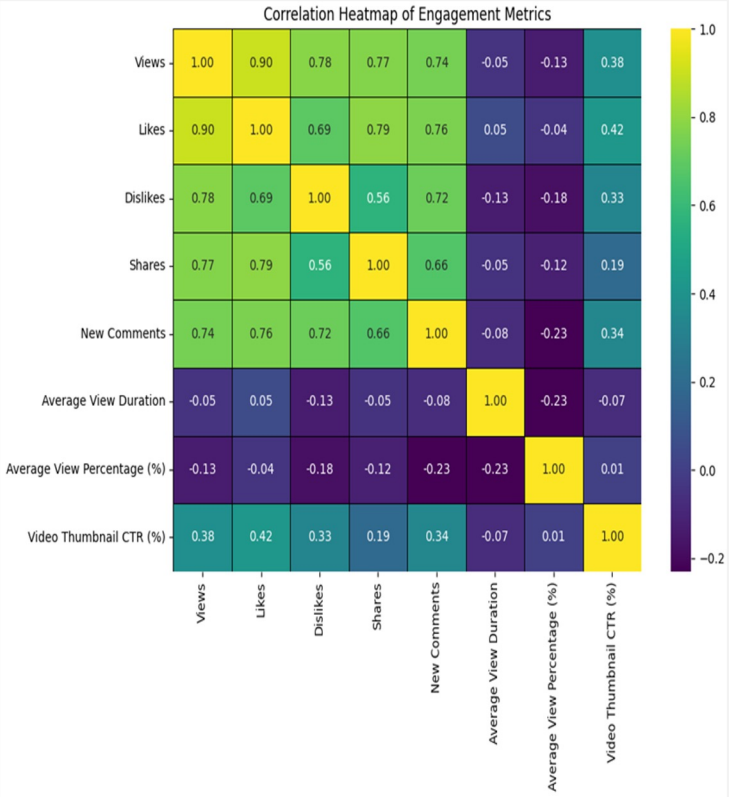
Engagement Metrics (Views vs. Likes, Comments, Shares)



Distribution of Video Duration



Correlation of Engagement Metrics



Modelling: Average View Percentage (AVP)

Target Variable

Average View Percentage (AVP) measures how much of a video is watched, on average, by viewers — a key indicator of content engagement and retention.

Project Goals

- Predict AVP using Linear Regression
- Identify key factors that influence viewer retention

Modeling Approach

- Removed features with high multicollinearity using VIF analysis
- Selected significant predictors ($p < 0.05$) to improve model interpretability
- Trained a Multiple Linear Regression model
- Evaluated model performance using R^2 and Mean Squared Error (MSE)
- Applied Ridge Regression with GridSearchCV for hyperparameter tuning
- Validated model performance using K-Fold Cross-Validation

Result: Average View Percentage (AVP)

Variable	Impact	Interpretation
Video Duration	Strong Negative	Longer videos significantly reduce viewer retention
Average View Duration	Strong Positive	Longer viewing sessions drive higher AVP
Playlist Views	Moderate Negative	Autoplay/skimming may lower retention
Like Rate (%)	Moderate Negative	High likes don't ensure longer watch time
Playlist Watch Time	Moderate Positive	Playlists encourage deeper engagement
Views per Playlist Start	Moderate Negative	Passive viewing reduces AVP
Year	Moderate Positive	Newer content performs better
Impressions	Small Positive	More exposure slightly boosts AVP
Dislikes	Small Negative	Higher dislikes are linked to lower engagement

OLS Model

- $R^2 = 0.788$
- Adjusted $R^2 = 0.781$
- MSE (Train): 14.36
- MSE (Test): 27.15
- R^2 (Test): 0.60

Ridge Model (after tuning)

- Cross-validated $R^2 \approx 0.735$
- MSE reduced to ~20.04

Practical Implication: Average View Percentage (AVP)

- Creators should focus on increasing average view duration rather than video length or likes.
- Short, structured, and engaging videos lead to better viewer retention.
- Passive playlist views often lower retention—encourage active viewing instead.
- Advertisers and brands can use these insights to target high-engagement content and improve Return on Investment (ROI).

Modelling: Revenue Per 1000 Views (RPM)

Target Variable

Revenue per 1,000 Views (RPM) measures how much a content creator earns for every 1,000 video views.

Project Goals

- Predict RPM using Linear Regression
- Identify key factors that influence revenue

Modeling Approach

- Selected statistically significant predictors ($p < 0.05$) for interpretability
- Trained a Multiple Linear Regression model to predict RPM
- Evaluated model performance using R^2 and Mean Squared Error (MSE)
- Applied Ridge Regression with GridSearchCV for hyperparameter tuning
- Validated model performance using K-Fold Cross-Validation

Result: Revenue Per 1000 Views (RPM)

Variable	Impact	Interpretation
Year	Strong Positive	Newer content is associated with higher RPM
Playback-Based CPM (USD)	Strong Positive	Higher advertiser payout per monetized playback increases RPM
YouTube Ads Revenue (USD)	Strong Positive	Nore total ad revenue directly increases RPM
CPM (USD))	Strong Negative	Longer engagement increases ad revenue potential
Likes	Moderate Negative	More likes slightly relate to lower RPM
Ad Impressions	Weak Negative	More impressions slightly reduce RPM
Playlist Watch Time (hours)	Weak Negative	More time spent on playlists slightly reduces RPM

OLS Model

- $R^2 = 0.976$
- Adjusted $R^2 = 0.974$
- MSE = 0.0022

Ridge Model (after tuning)

- Cross-validated $R^2 \approx 0.784$
- MSE = 0.0022

Practical Implication: Revenue per 1000 (RPM)

- Higher Playback-Based CPM and YouTube Ads Revenue strongly increase Revenue per 1000 Views, highlighting the importance of ad monetization quality.
- Metrics like CPM (USD), Likes, and Playlist Watch Time negatively impact RPM, suggesting that engagement doesn't always translate to higher earnings.
- With an Adjusted R^2 of 0.794, the model explains 79% of the variation in RPM, making it a strong tool for revenue optimization decisions.
- To improve RPM, YouTubers should focus on boosting Playback-Based CPM and Ads Revenue.

Modeling: Viewer Retention Metrics

Target Variable

The viewer retention metrics will measure what variables are important when trying to keep viewer retention

Project Goals

- Predict retention using Linear Regression
- Identify key factors that influence viewer retention

Modeling Approach

- Conduct correlation analysis to identify top predictors associated with retention and remove insignificant variables
- Trained a Multilinear Regression Model and used Random Forest Regressor to improve prediction accuracy
- Evaluated model performance using R^2 and Mean Squared Error (MSE)
- Plotted residual distribution data

Result: Viewer Retention Metrics

Variable	Impact	Interpretation
Video Duration	Strong Positive	Longer videos drive more watch time
Impressions	Strong Positive	Higher impressions increase visibility
Shares	Strong Negative	Highly shared videos might be short or click worthy but not engaging
Like Rate	Moderate Negative	High like rate may not translate to long watch time
Video Thumbnail CTR	Weak Negative	Interesting thumbnails may draw clicks but not longer retention
New Subscribers	Weak Negative	Videos that gain new subscribers may not be the ones with high watch times

OLS Model

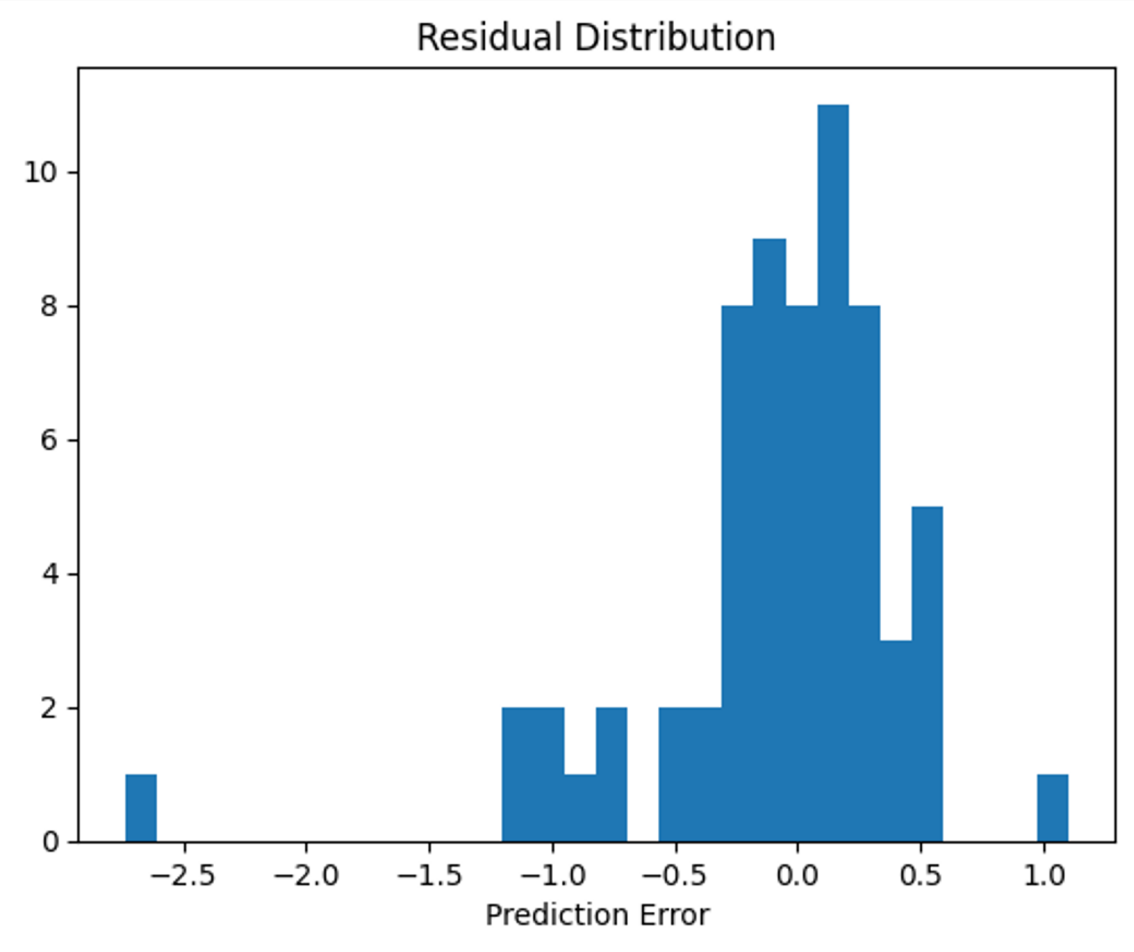
- $R^2 = 0.799$
- Adjusted $R^2 = 0.794$
- Training MSE = 0.157
- Training $R^2 = 0.904$
- Test MSE = 0.297
- Test $R^2 = 0.860$

Practical Implication: Viewer Retention

- Content creators should focus on making longer form content that engages the audience
- Boosting visibility through optimized titles, thumbnails, and search engine optimization (SEO) increases impressions and views
- Like Rates and New Subscribers are not direct indicators of retention

Residual Distribution Histogram

- Shows prediction errors (y_test - y_pred)
- Bell shaped curve with a symmetric spread
- No strong skew



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Thank You!